DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE		
Rescheduled Supplementary Summer Examination – 2025		
Course: B. Tech.	**Branch : Computer Engineering / Computer Science and Engineering**	**Semester :V**
Subject Code & Name: BTCOC502 & Theory of Computation		
Max Marks: 60	**Date: 25**/08/2025	Duration: 3Hr.

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

					(Level/CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**					**12**
1	Which of the following is not a part of tuple finite automata?				Understand	1
	a.Input alphabet	b.Transition function	c.Initial State	d. Output function		
2	The number of states required by an automaton to accept string 101 are				Analyze	1
	a. 2	b. 3	c.4	d.5		
3	Regular expression for string that start with ab and end with ba is				Design	1
	a. aba*b*ba	b. ab(ab)*ba	c. ab(a+b)*ba	d. All of these		
4	Regular expressions are closed under				Understand	1
	a. Union	b.Concatenation	c. Kleene Star	d. All of these		
5	Which grammar is in Greibach normal form (GNF) ?				Understand	1
	a. $S \rightarrow AA\|0$, $A \rightarrow SS\ \|1$	b. $A \rightarrow aB$ $B \rightarrow b$	c. $A \rightarrow aBd$ $B \rightarrow b$	d. None of these		
6	Moore machine produces output over the change of				Understand	1
	a. transitions	b. state	c. input symbol	d. None of these		

7	Can Deterministic Finite automata determine palindrome number ?				Analyze	1
	a. Yes	b. NO	c.Yes for Some cases	d. yes for Z*		
8	A grammar that produces more than one parse tree is called				Understand	1
	a. regular	b. ambiguous	c.unambiguous	d. linear		
9	The language accepted by Pushdown automata is				Understand	1
	a. type 0	b. type 1	c. type 2	d. type 3		
10	The language accepted by Turning Machine is				Understand	1
	a. type 0	b. type 1	c. type 2	d. type 3		
11	The Pushdown automata behaves like Turing machine if it has auxiliary memory				Understand	1
	a.0	b. 2	c.2 or more	d. None		
12	The language accepted by Finite Automata is				Understand	1
	a. type 0	b. type 1	c. type 2	d. type 3		
Q. 2	**Solve the following.**					**12**
A)	Find the string set for following regular expressions. (i) 00* (ii) a*b* (iii) (0+1)*				Create & CO1	**6**
B)	What is FA(Finite Automaton)? Explain with examples. Elaborate on Automaton and complexity				Understand & CO1	**6**
Q.3	**Solve the following.**					**12**
A)	Explain in brief Chomsky hierarchy with suitable examples?				Understand & CO2	**6**
B)	Design a FSM to check whether a given decimal number is divisible by 5?				Understand & CO1	**6**
Q. 4	**Solve Any Two of the following.**					**12**
A)	Eliminate NULL production from the grammar G given below: Write the productions after elimination a. S→ a / Xb / aYa b. X→ Y / ^ c. Y→ b / X				Apply & CO2	**6**

B)	Consider a left linear grammar as given below a. S⟶ Sa|Abc b. A⟶ Sa|Ab|a Convert left linear grammar to right linear grammar.	Apply & CO3	**6**
C)	Convert the given to context free grammar to CNF forms a. S ⟶ ASB b. A ⟶ aAS|a|ε c. B ⟶ SbS|A|bb	Apply & CO-3	6
Q.5	**Solve Any Two of the following.**		**12**
A)	Derive a derivation tree for the string aabbabba for the given context free grammar (CFG) a. S->aB|bA b. A->a|aS|bAA c. B->b|bS|aBB	Analyse / CO-3	**6**
B)	Define a PDA & list three important properties of a PDA Machine?	Understand & CO-2	6
C)	Convert the following grammar to a PDA that accepts the same language. a. S ⟶ 0S1 | A b. A ⟶ 1A0 | S | ε	Apply & CO-3	**6**
Q. 6	**Solve Any Two of the following.**		**12**
A)	Design a TM for an equal number of a's and b's must follow a.	Create & CO-3	**6**
B)	Design a TM which recognizes palindromes over = {a, b}	Create & CO-3	**6**
C)	Explain 'Halting Problem of turning machine' with neat diagrams?	Understand & CO-4	**6**
	*** **End** ***		